

# Do We Need 3D to See? Impact of Dimensionality of the Virtual Environment on Attention

Teresa Matos<sup>\*1,3</sup>, Daniel Mendes<sup>†1,2</sup>, João Jacob<sup>‡3</sup>, A. Augusto de Sousa<sup>§1,2</sup>, and Rui Rodrigues<sup>¶1,2</sup>

<sup>1</sup>Faculdade de Engenharia, Universidade do Porto, <sup>2</sup>INESC TEC, <sup>3</sup>Mindera

## ABSTRACT

Virtual Reality allows users to experience realistic environments in an immersive and controlled manner, particularly beneficial for contexts where the real scenario is not easily or safely accessible. The choice between 360° content and 3D models impacts outcomes such as perceived quality and computational cost, but can also affect user attention. This study explores how attention manifests in VR using a 3D model or a 360° image rendered from said model during visuospatial tasks. User tests revealed no significant difference in workload or cybersickness between these types of content, while sense of presence was reportedly higher in the 3D environment.

**Index Terms:** Virtual reality, visual attention, task performance

## 1 INTRODUCTION

Virtual Reality places users in an immersive and virtual environment, giving them the feeling of being there. It allows them to interact with that environment similarly to how they would in the real world, which can be used to create or change behaviours that are then transferable to real situations [1]. The benefits of this spread to several applications, such as VR training and learning. In these contexts, a vital component is visual attention, as users have to perform tasks which require their focus on different parts of their surroundings during the experience [15]. Creating effective experiences requires balancing realistic environment representation with designing areas of interest that are easy to detect and focus on [8].

A real environment can be simulated in a VR experience using either 2D assets, such as 360° images/videos captured from said environment, or instead a 3D model of it, which can be created with 3D modelling tools or using techniques such as photogrammetry. Creating 360° can be easier and faster, although it can provide a more limiting experience due to lack of depth data and restrictions over user movement (3DoF). Alternatively, 3D models allow for more interactive experiences with higher quality, yet potentially higher production and computational cost. This choice can be impactful on development time and cost but also on desired outcomes of the experience. Previous works have looked at its impact on sense of presence, cybersickness, engagement, among others [2]. Beyond these concepts, it can influence users' attention, which is essential while performing tasks and retaining information [3].

With this work, we examined the potential connection between attention and the virtual environment (VE), aiming to support future decisions for creating more effective and engaging immersive experiences. We defined two research questions: (RQ1) "Does the

dimensionality of an object's representation affect the performance of visuospatial tasks in VR?" and (RQ2) "Does an object's dimensionality affect its visual saliency?". We performed user studies to analyse how users' attention manifests during visuospatial tasks in a VE presented as a 3D model or a monoscopic 360° image. Results gathered from an eye-tracking system and questionnaires on workload, presence and cybersickness overall showed no significant difference between these two types of content, suggesting that this may not be a deciding factor for VR in the explored contexts.

## 2 METHODOLOGY

The user studies (n=22) consisted of performing visuospatial tasks with a VR headset in a scenario presented as a 3D model or a monoscopic 360° image of said model. Based on previous related work [5], the following tasks were selected: **FIND** - Find a specific object; **COUNT** - Count the number of objects of a specific type; and **SALIENCY** - Report the most salient element or region. These tasks were chosen for being compatible with a static environment where user movement is not possible (due to the 360° image). Participants performed the 3 tasks in this order in both environment conditions, with alternating targets (i.e., the objects to count or find) and positions for each condition. The order of the environment and target versions were counterbalanced to minimize learning effects.

A prototype in Unity was developed for the user studies, and a HTC Vive headset was used with the Pupil Labs HTC Vive Add-on for eye-tracking features. For the virtual environment, the Amazon Lumberryard Bistro model was used [9], where the interior of the Bistro was used for the task phase, and the exterior was presented during calibration and between tasks, as shown in Fig. 1. The 360° images were captured from the 3D model using Unity Recorder package, so that both conditions were of similar visual quality with consistent lighting and material properties [14].

In each test session, the participant was requested to fill out a demographics questionnaire, followed by a calibration phase of the eye-tracking system. Once the calibration was successful, the participants would start performing the tasks. The raw NASA-TLX questionnaire [4] was filled after tasks FIND and COUNT, and the



Figure 1: Selected environment for the user studies [9]. Scene for the task phase (left); Scene for calibration and in-between tasks (right).

\*e-mail: up201105540@up.pt

†e-mail: danielmendes@fe.up.pt

‡e-mail: joao.jacob@mindera.com

§e-mail: augusto.sousa@fe.up.pt

¶e-mail: rui.rodrigues@fe.up.pt

IGroup Presence Questionnaire (IPQ) [13] together with the Simulator Sickness Questionnaire (SSQ) [6] were filled after each condition. At the end of the session, the two conditions were compared directly in a final questionnaire. Each task was timed, and eye-tracking data was recorded: pupil changes, blinks, and fixations.

### 3 RESULTS

User tests' data showed, for the most part, no significant differences between the two conditions. This section details the results, together with graphs for the quantitative data in Fig. 2, to address our research questions and draw conclusions. Participants reported a lower **workload** in the 3D environment, but not to a significant degree for tasks FIND ( $Z = -1.73$ ,  $p = 0.08$ ) and COUNT ( $Z = -0.76$ ,  $p = 0.45$ ). Overall, participants felt more **present** in the 3D model condition ( $Z = -2.87$ ,  $p < 0.01$ ), to a significant degree for all subscales. Most users reported no cybersickness symptoms during the experience (avg. test duration of 44 min.), as expected in static VR experiences without user locomotion. In the end-session questionnaire (5-point scale), users rated a noticeable visual difference between conditions (Mdn = 4.5, IQR = 2) and showed a clear preference for the 3D scene over the 360° image (Mdn = 4, IQR = 1). Lastly, users found it easier to discern objects in the 3D condition (Mdn = 2, IQR = 2). Wilcoxon signed-rank tests showed significant differences for all 3 questions compared to the neutral preference.

Users took more **time** for the FIND task but less for the COUNT task in the 3D condition, neither to a significant degree (FIND:  $Z = -0.31$ ,  $p = 0.76$ ; COUNT:  $Z = -0.92$ ,  $p = 0.36$ ). From the eye-tracking data collected during the user tests, **pupil diameter** changes were evaluated relative to a baseline value for each user calculated before the task phase. The 3D condition led to smaller pupil diameter values, which could point at lower cognitive load [7], however no significant difference was found for any of the tasks. **Blink rates** were calculated during each task and compared to a baseline rate, also established before the tasks started. Blink rates were inferior (though not significant) in the 3D condition. This could point at a higher cognitive load, although considering the data from blink rates, it could be attributed to higher level of engagement [10]. **Fixations** are detected by the Pupil software, following the work by Salvucci and Goldberg [12]. The mean fixation duration (MFD) was on average shorter in the 3D environment, with statistical difference for task COUNT ( $Z = -2.59$ ,  $p = 0.01$ ) but not for task FIND

( $Z = -1.25$ ,  $p = 0.21$ ). Longer fixation durations in the 360° environment could be indicative of difficulty in extracting details from the environment during the visual search tasks [11]. Finally, regarding the **SALIENCY** task, the most common salient characteristics for the 360° condition were lighting (28.6%) followed by colour (23.8%), and visual richness (14.3%) while in the 3D condition, these were lighting (33.3%), proximity and geometry (23.8% each). Most users (71.4%) changed their answers between the two conditions to a significant degree ( $\chi^2(2) = -2.35$ ,  $p = 0.02$ ).

### 4 CONCLUSION

For VR experiences aiming to immerse users in a real-world environment to adopt or train behaviours, an important question is whether to use 360° content or 3D models. While prior studies have focused on factors such as presence or emotion, this work examines the impact on user attention. Preliminary tests indicate no significant difference in performing visuospatial tasks, such as locating or distinguishing objects, between 3D and 360° image representations (RQ1). Participant feedback highlights that depth cues in 3D models can be moderately helpful for distant or similarly coloured elements. Additionally, object saliency differed significantly between conditions, with depth features being more prominent in 3D scenes (RQ2). Future studies should compare different VEs and include other tasks for more conclusive and generalizable findings, as well as study stereoscopic 360° images which may shed light on how perceived depth affects attention in VR. This research could support the VE design process and help create more cost-efficient VR experiences, without hindering sense of presence or user attention.

### REFERENCES

- [1] C. Bossard et al. Transfer of learning in virtual environments: A new challenge? *Virtual Reality*, 12(3), 2008.
- [2] M. Boukhris et al. *360° versus 3D Environments in VR Headsets for an Exploration Task*. The Eurographics Association, 2017.
- [3] N. Cowan et al. The Relation Between Attention and Memory. *Annual Review of Psychology*, 75(1), 2024.
- [4] S. G. Hart. Nasa-Task Load Index (NASA-TLX); 20 Years Later. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, 50(9), 2006.
- [5] Z. Hu et al. Ehtask: Recognizing user tasks from eye and head movements in immersive virtual reality. *IEEE Transactions on Visualization and Computer Graphics*, 29(4), 2021.
- [6] R. S. Kennedy et al. Simulator Sickness Questionnaire: An Enhanced Method for Quantifying Simulator Sickness. *The International Journal of Aviation Psychology*, 3(3), 1993.
- [7] H. Liu et al. Does stereopsis improve face identification? A study using a virtual reality display with integrated eye-tracking and pupilometry. *Acta Psychologica*, 210, 2020.
- [8] T. Matos et al. Dynamic annotations on an interactive web-based 360° video player. In *Proceedings of the 23rd International ACM Conference on 3D Web Technology*. ACM, 2018.
- [9] M. McGuire. Computer graphics archive, 2017.
- [10] J. Moreno-Arjonilla et al. Eye-tracking on virtual reality: A survey. *Virtual Reality*, 28(1), 2024.
- [11] A. Poole and L. J. Ball. Eye Tracking in HCI and Usability Research. *Encyclopedia Of Human Computer Interaction*, 2006.
- [12] D. D. Salvucci and J. H. Goldberg. Identifying fixations and saccades in eye-tracking protocols. In *Proceedings of the 2000 symposium on Eye tracking research & applications*, 2000.
- [13] T. Schubert et al. The Experience of Presence: Factor Analytic Insights. *Presence: Teleoperators and Virtual Environments*, 10(3), 2001.
- [14] T. Teo et al. Mixed reality remote collaboration combining 360 video and 3d reconstruction. In *Proceedings of the 2019 CHI conference on human factors in computing systems*, 2019.
- [15] Y. Valentin and H. Choi. Visual attention in extended reality and implications for aviation safety. In *International Conference on Human-Computer Interaction*. Springer, 2023.

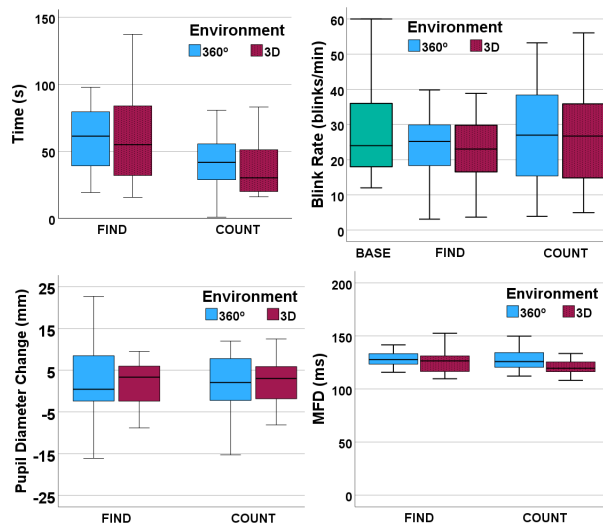


Figure 2: Boxplots for time (top-left), blink rate (top-right), pupil diameter changes (bottom-left) and mean fixation duration (bottom-right).